EMIPredict AI – Exploratory Data Analysis Report

Intelligent Financial Risk Assessment Platform – FinTech and Banking

Live app: https://abhisheknimaje435-emipredict-ai-app-ccio5a.streamlit.app

Repository: https://github.com/AbhishekNimaje435/emipredict-ai

1. Introduction and objectives

This report summarises the exploratory analysis of the EMI dataset used by EMIPredict AI. The aim is to understand data quality, the distribution of the eligibility target, and the financial and demographic factors that drive EMI eligibility and the maximum safe monthly EMI, before building the classification and regression models.

Objectives (as per the project brief): analyse eligibility distribution across lending scenarios, study correlations between financial variables and approval, investigate demographic patterns and risk factors, and produce statistical summaries with business insights.

2. Dataset overview

Item	Value
Raw file	404,800 rows × 27 columns (25 input features + 2 targets)
After cleaning	400,024 rows
EMI scenarios	E-commerce Shopping, Home Appliances, Vehicle, Personal Loan, Education (about 80,000 records each)
Classification target	emi_eligibility: Eligible, High_Risk, Not_Eligible
Regression target	max_monthly_emi (INR), from near zero up to about ₹90,000 (see Section 4.2)
Split (stratified 70/15/15)	Train 280,016 / Validation 60,004 / Test 60,004

Feature groups

Group	Variables
Personal demographics	age, gender, marital_status, education
Employment and income	monthly_salary, employment_type, years_of_employment, company_type
Housing and family	house_type, monthly_rent, family_size, dependents
Monthly obligations	school_fees, college_fees, travel_expenses, groceries_utilities, other_monthly_expenses
Financial status	existing_loans, current_emi_amount, credit_score, bank_balance, emergency_fund
Loan application	emi_scenario, requested_amount, requested_tenure

Profile of the cleaned numeric features

Feature	Minimum	Median	Maximum
age (years)	26	38	59
monthly_salary (₹)	12,000	51,700	175,900
years_of_employment	0.5	3.2	36
credit_score	407	701	850
bank_balance (₹)	19,600	195,900	845,600
emergency_fund (₹)	6,600	74,000	384,000
current_emi_amount (₹)	0	0	56,300
requested_amount (₹)	20,000	236,000	1,430,000
requested_tenure (months)	3	25	84

The median existing EMI is zero, so more than half of the applicants carry no current EMI. The salary range starts at about ₹12,000 and ends at about ₹175,900 because extreme values were clipped to the 1st–99th percentile during preprocessing.

3. Data quality assessment

The quality check on the raw data found the issues below. Each was handled in step1_preprocessing.py.

Issue found	Treatment
Missing values: education (2,404), monthly_rent (2,426), credit_score (2,420), bank_balance (2,426), emergency_fund (2,351)	Median imputation (numeric), mode imputation (categorical)
age, monthly_salary and bank_balance stored as text because of noisy entries	Non-numeric characters stripped, converted to numbers
Inconsistent gender labels: F, FEMALE, Female, female, M, MALE, Male, male	Normalised to Female and Male
Invalid credit_score values (above the 300–850 range, up to 1200)	Rows outside 300–850 removed (about 4,800 rows, 1.2%)
Extreme values in salary, balances and requested amount	Clipped to the 1st–99th percentile
Duplicate rows	None found

Insight: Missing values are small (each about 0.6% of rows or less) and spread evenly across five columns, so median/mode imputation introduces very little bias. Removing invalid credit scores cost only about 1.2% of the data (404,800 raw rows became 400,024).

4. Target distribution

4.1 EMI eligibility (classification target)

Class	Share	Meaning
Not_Eligible	77.3%	High risk, loan not recommended
Eligible	18.4%	Low risk, comfortable affordability
High_Risk	4.3%	Marginal case, needs higher interest rate

Insight: The data is strongly imbalanced. A model that always predicts Not_Eligible would already reach about 77% accuracy, so accuracy alone is misleading. This is why balanced sample weights and macro F1 were used for model selection, and why High_Risk recall is tracked separately.

4.2 Maximum monthly EMI (regression target)

The distribution plot of max_monthly_emi shows values from near zero up to about ₹90,000. The distribution is heavily right-skewed: a very large group of applicants (roughly 140,000 records, about a third of the data) falls in the lowest EMI bins, and the counts then fall steadily towards a long tail. Across the five scenarios the average maximum EMI is about ₹6,760.

Most applicants therefore have a small safe EMI, and only a few have a high one. This also explains why MAPE (7.26% on the test set) is much higher than the RMSE would suggest: small target values make percentage errors large. A log transform of the target could be tried to reduce the skew in future work.

5. Eligibility across EMI scenarios

Goal: compare how eligibility classes are distributed across the five lending scenarios.

Scenario	Eligible %	Not_Eligible %	High_Risk %	Mean max EMI (₹)
E-commerce Shopping EMI	26.7	68.4	4.9	6,783

Scenario	Eligible %	Not_Eligible %	High_Risk %	Mean max EMI (₹)
Home Appliances EMI	25.4	69.5	5.1	6,759
Education EMI	18.2	77.3	4.5	6,754
Personal Loan EMI	11.7	84.6	3.7	6,735
Vehicle EMI	10.7	85.9	3.4	6,778

The class percentages are computed on the 20,000-record sample used by the app; the mean max EMI values are from the full cleaned dataset. Vehicle (85.9%) and Personal Loan (84.6%) have the highest rejection rates, and E-commerce Shopping (68.4%) the lowest.

Insight 1: The average maximum safe EMI is almost the same in every scenario (about ₹6,735 to ₹6,783, a spread of under 1%). The loan type itself barely changes how much EMI an applicant can afford; affordability is driven by the applicant's own finances.

Insight 2: Eligibility nevertheless differs sharply by scenario. Smaller-ticket, shorter-tenure products (E-commerce, Home Appliances) have the highest eligibility, and high-ticket products (Vehicle, Personal Loan) the lowest. The reason is not a different safe EMI but a larger requested EMI, which consumes more of the applicant's disposable income.

6. Financial variables vs eligibility

6.1 Correlation with maximum monthly EMI

Variable	Correlation with max_monthly_emi
monthly_salary	0.53
groceries_utilities	0.48
bank_balance	0.45
travel_expenses	0.44
emergency_fund	0.41
other_monthly_expenses	0.38

Variable	Correlation with max_monthly_emi
credit_score	0.30
years_of_employment	0.03

Insights: Monthly salary is the strongest single driver (0.53), followed by bank balance (0.45) and emergency fund (0.41), so both income and savings matter. Credit score has a moderate link (0.30). Years of employment is almost unrelated (0.03). Spending columns such as groceries_utilities (0.48) and travel_expenses (0.44) also correlate positively; most likely because people with higher income also spend more, so these columns act as indirect income signals. They do not mean that spending raises the safe EMI.

6.2 Credit score by eligibility class

Class	Mean credit score
Eligible	725.5
High_Risk	715.9
Not_Eligible	693.6

Insight: The gap between the best and worst group is about 32 points. Credit score separates the classes, but only moderately, so it cannot decide eligibility alone; the debt and income ratios must matter too.

6.3 Salary vs eligibility and vs maximum EMI

Statistic (approximate, read from the box plot)	Eligible	High_Risk	Not_Eligible
Median monthly salary (₹)	~72,000	~63,000	~47,000

Insight: The median salary is highest for Eligible applicants, then High_Risk, and lowest for Not_Eligible. Not_Eligible still contains many high-salary outliers up to the cap of about ₹175,000, so a high salary does not guarantee eligibility; obligations, existing EMIs and savings also decide the outcome.

In the salary vs max EMI scatter plot, max EMI rises with salary but the spread at any salary level is wide, so salary alone cannot predict it. Eligible applicants mostly sit in the upper part of the cloud, while Not_Eligible applicants crowd near the bottom, many close to zero EMI even at medium and high salary. The vertical lines of points at the right edge (about ₹175,000) and the left edge (about ₹12,000) are caused by clipping salary to the 1st–99th percentile in preprocessing.

The model results show the same non-linearity: Linear Regression (R^2 0.79) is far behind tree-based models (R^2 up to 0.99), which indicates that max EMI depends on non-linear interactions between income, obligations and credit profile.

7. Demographic patterns and fairness

The demographic variables in the data are age (26 to 59 years, median 38), gender (Female, Male), marital status (Married, Single), education (High School, Graduate, Post Graduate, Professional), employment type (Government, Private, Self-employed), company type (Large Indian, MNC, Mid-size, Small, Startup), house type (Own, Rented, Family), family size (1 to 5) and dependents (0 to 4).

Eligibility differs little by age and gender but strongly by education, housing and dependents. The table shows the share of each group in every class (20,000-record sample).

Variable	Group	Eligible %	High_Risk %	Not_Eligible %
Age band	26–35	18.3	4.2	77.5
Age band	36–45	18.7	4.4	76.9
Age band	46–59	18.7	4.4	77.0
Gender	Female	18.9	4.3	76.8
Gender	Male	18.3	4.3	77.4
Marital status	Married	17.9	4.1	78.0
Marital status	Single	20.6	5.0	74.4
Education	High School	11.2	3.5	85.3
Education	Graduate	15.9	4.2	79.9
Education	Post Graduate	22.4	4.2	73.3

Variable	Group	Eligible %	High_Risk %	Not_Eligible %
Education	Professional	27.6	5.7	66.7
Employment type	Government	22.3	4.2	73.5
Employment type	Private	17.7	4.4	78.0
Employment type	Self-employed	17.2	4.4	78.4
House type	Own	24.9	5.6	69.5
House type	Family	23.9	5.1	71.0
House type	Rented	9.7	2.7	87.7
Dependents	0	25.3	6.6	68.1
Dependents	1	18.4	3.9	77.7
Dependents	2	17.7	4.1	78.2
Dependents	3	17.4	4.3	78.2
Dependents	4	18.4	3.9	77.8

Insight 1: Age is almost irrelevant. The Eligible share stays between 18.3% and 18.7% in all three age bands.

Insight 2: Gender differences are small (Female 18.9% vs Male 18.3% eligible, a 0.6-point gap). This agrees with gender and marital status not appearing in the top 15 features of either model. Single applicants are eligible somewhat more often than married ones (20.6% vs 17.9%).

Insight 3: Education shows the clearest gradient. Eligibility rises from 11.2% (High School) through 15.9% (Graduate) and 22.4% (Post Graduate) to 27.6% (Professional). This most likely reflects income differences, but it was not tested here.

Insight 4: Housing matters. Renters are eligible only 9.7% of the time, against 24.9% for owners and 23.9% for family housing. This is consistent with house_type Rented appearing in the regressor's top 15 features, since rent is part of total expenses and lowers disposable income.

Insight 5: Applicants with no dependents are eligible 25.3% of the time; with 1 to 4 dependents the rate falls to about 17% to 18% and stays flat. Government employees are eligible more often (22.3%) than private (17.7%) and self-employed (17.2%) applicants.

A formal fairness audit (tests on real lending data and a check of proxy variables such as education and housing) was not part of this project. The gaps above are descriptive only, and such an audit is recommended before any real-world use.

8. Engineered features

The following features are created by add_features() and used identically in training and in the live app:

Type	Features
Financial ratios	total expenses, debt-to-income, expense-to-income, disposable income, affordability ratio, requested EMI estimate (amount / tenure), EMI-to-income, EMI-to-disposable-income, loan-to-annual-income
Savings and risk	savings buffer (months), emergency-fund cover, credit risk score (0–1), employment stability, dependents ratio
Interactions	salary × credit risk score, debt-to-income × (1 – credit risk score), EMI burden × dependents

These 17 engineered features plus the original inputs give 34 numeric columns; after one-hot encoding of the 8 categorical columns (26 columns) and scaling, both models use 60 input features. Encoding and scaling were fitted on the training split only.

8.1 What the selected XGBoost models rely on

Classifier – approximate importance values read from the top-15 feature importance chart:

Feature	Importance (approx.)
emi_to_disposable	0.16
existing_loans (Yes)	0.09
existing_loans (No)	0.09
dti_x_credit	0.07

Feature	Importance (approx.)
emi_to_income	0.05
affordability_ratio	0.03
monthly_salary (raw)	0.02

Insight (classifier): The most important single feature is emi_to_disposable, an engineered ratio. existing_loans appears twice because of one-hot encoding; the two columns are complementary, so together they carry about 0.18, roughly the same weight as the top ratio. The next features are also engineered. Raw monthly_salary has only about 0.02. This supports the feature-engineering step: ratios that compare the EMI to what the person can afford carry more information than raw amounts.

Regressor:

Feature	Importance (approx.)
existing_loans (Yes)	0.65
disposable_income	0.14
every other feature	below 0.03

Insight (regressor): The predicted maximum EMI depends mostly on whether the applicant already has loans and on how much income is left after expenses. This differs from the correlation view, where salary was the top correlate (0.53), because correlation measures only linear links while trees can split on a yes/no variable. Feature importance shows what the model uses, not what causes the outcome, and such a strong dependence on one variable is worth checking against how the target was calculated.

8.2 Classifier confusion matrix (validation set, 60,004 records)

Actual \ Predicted	Eligible	High_Risk	Not_Eligible
Eligible	10,588	385	60
High_Risk	128	2,348	117
Not_Eligible	69	727	45,582

Insight: 58,518 of 60,004 records are classified correctly (97.5%). Per-class recall is Eligible 96.0%, High_Risk 90.6% and Not_Eligible 98.3%. Mistakes are mostly between neighbouring classes: only 60 Eligible applicants were predicted Not_Eligible and 69 Not_Eligible applicants were predicted Eligible. The largest error is 727 Not_Eligible applicants predicted High_Risk, which brings High_Risk precision down to about 68% (2,348 of 3,460 predictions). In practice the model rarely makes a costly wrong approval, but it flags some extra applicants as marginal, which is a safe direction for a lender.

9. Key findings and business insights

1. About 77% of applicants are Not_Eligible and only 4.3% are High_Risk. Lenders should expect mostly rejections, and the marginal group is small but important for risk-based pricing.
2. Data quality was good: only about 1.2% of rows were removed and no duplicates were found, so conclusions are not driven by data errors.
3. Non-linear patterns dominate: tree-based models far outperform linear ones for both tasks, so simple cut-offs on single variables are not enough.
4. The High_Risk class can be identified well (about 90% recall) despite its small share, supporting a three-tier decision (approve / price higher / reject).
5. Loan scenario does not change affordability: average max EMI is about ₹6,735 to ₹6,783 in all five scenarios. Eligibility still differs by scenario because the requested EMI differs.
6. Salary, bank balance and emergency fund are the strongest correlates of safe EMI (0.53, 0.45, 0.41), so lenders should assess income and savings together, not income alone.
7. Credit score matters but only moderately: 725.5 (Eligible) vs 693.6 (Not_Eligible) on average. It should be one input among several.
8. Years of employment has almost no linear link with max EMI (0.03).
9. Max EMI is strongly right-skewed: most applicants can afford only a small EMI, and few can afford a large one.
10. High salary alone does not secure eligibility: Not_Eligible applicants include many high earners.
11. Engineered ratios (EMI-to-disposable income, debt-to-income × credit, EMI-to-income) are the strongest predictors of eligibility, and existing loans plus disposable income drive the maximum EMI.

12. Direct Eligible/Not_Eligible mix-ups are very rare (60 and 69 of 60,004); most errors involve the High_Risk class.

10. Recommendations

1. Use macro F1 and per-class recall, not accuracy, to monitor models because of class imbalance.
2. Keep strict input validation (credit score 300–850, consistent categorical labels) at data entry to avoid the quality issues seen in the raw data.
3. Review model outcomes across gender and other demographic groups before any production use.
4. Use the predicted High_Risk class for risk-based pricing instead of a simple approve/reject rule.
5. Consider a log transform of max_monthly_emi to handle its heavy right skew.

This is an educational project. Its predictions are not a substitute for real lending decisions.